

Chapter 3- CPU, RAM, GPU & System Information

Processor (CPU – Central Processing Unit)

★ What is CPU?

- 👉 CPU is the **brain of the computer**
- 👉 It performs all calculations and processing

Cores & Threads (Very Important)

◆ Core:

- 👉 A **physical unit** inside CPU that performs tasks

◆ Thread:

- 👉 A **virtual processing unit** (logical path)

Concept:

- 👉 1 Core = Can handle 1 task
- 👉 1 Core + 2 Threads = Can handle **2 tasks simultaneously**

Example:

- 👉 Like a worker:

- Core = Worker 
- Thread = Hands 

- 👉 More cores = More workers → Faster work
-

Why More Cores?

 Enables **multitasking & multiprocessing**

CPU Categories (General Use)

CPU Use

- i3 Basic work (browsing, office)
 - i5 Medium tasks (students, coding)
 - i7 Heavy tasks (gaming, editing)
 - i9 Extreme tasks (servers, streaming)
-

Intel CPU Generations (Latest → Oldest)

Latest

- 15th Gen – Arrow Lake (AI + efficiency)
- 14th Gen – Raptor Lake Refresh
- 13th Gen – Raptor Lake
- 12th Gen – Alder Lake (P + E cores)

Mid

- 11th Gen – Rocket Lake
- 10th Gen – Comet Lake
- 9th Gen – Coffee Lake Refresh
- 8th Gen – Coffee Lake

Older

- 7th – Kaby Lake
- 6th – Skylake
- 5th – Broadwell

- 4th – Haswell

● Legacy

- 3rd – Ivy Bridge
 - 2nd – Sandy Bridge
 - 1st – Nehalem
-

◆ AMD CPU Generations

● Latest

- Ryzen 8000 Series
- Ryzen 7000 Series
- Ryzen 5000 Series

● Mid

- Ryzen 3000
- Ryzen 2000
- Ryzen 1000

● Old

- FX Series
 - Phenom II
-

🍏 Apple Silicon (Mac Processors)

● Latest

- M5 Series
- M4 Series
- M3 Series

● Mid

- M2 Series
- M1 Series

● Old

- Intel-based Macs
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🔥 Motherboard & Processor Heating Concept

👉 Inside motherboard:

- Copper traces (wires) carry current ⚡
- Current creates **electromagnetic fields**

👉 Processor gets power through **VRM coils**

⚠️ Important:

👉 More power → More heat

💡 Practical Tip:

❌ Do NOT use laptop on:

- Bed 🛏️
- Blanket

👉 Why?

- Blocks airflow
 - Dust enters fan
 - Overheating occurs
-

✅ **Correct Use:**

👉 Keep laptop on flat surface

👉 Maintain airflow 🌬️



System Information Commands (VERY IMPORTANT)



1 msinfo32

How:

👉 Press **Windows + R** → type **msinfo32**

Shows:

- OS version
 - Processor
 - RAM
 - Motherboard
 - BIOS mode
-



2 winver

👉 Shows Windows version



CPU Naming Understanding (Important for Interview)

Example:

👉 i5-8250U

- 8 → 8th Generation
 - U → Low power CPU
-



CPU Suffix Meaning

Suffix Meaning

U / Y Low power (battery saving)

P Balanced

H High performance

HX Extreme performance

K Overclockable

F No GPU



RAM (Memory)



What is RAM?



Temporary memory used by CPU



Function:



Stores data for running applications



Example:



RAM = Study table



CPU = Student



RAM Types (Newest → Oldest)

Type Description

DDR5 Fastest

DDR4 Most common

DDR3 Old

DDR2 Very old



Concept:



DDR = Double Data Rate



Means faster data transfer



Mobile Processors



Latest

- Snapdragon 8 Gen 4
- Apple A18 Pro
- Dimensity 9400



Mid

- Snapdragon 8 Gen 3
- A17 Pro



Older

- Snapdragon 888
- A15 Bionic



Android Versions



Latest

- Android 16
- Android 17 (Upcoming)



Mid

- Android 15, 14



Old

- Android 13 and below



Check in Phone:



Settings → About Phone



GPU (Graphics Processing Unit)



What is GPU?



Handles **graphics, video, pixels**



CPU vs GPU:

CPU	GPU
Processes logic	Handles display
General tasks	Graphics tasks



Example:



CPU = Brain



GPU = Artist 🎨

Use Cases:

- Gaming
 - Video editing
 - Streaming
-

NVIDIA GPUs (Latest → Old)

Latest:

- RTX 5090
- RTX 50 Series

Mid:

- RTX 40 Series

Older:

- RTX 30 / 20

Old:

- GTX Series
-

Troubleshooting Concepts

System Slow Problem

Scenario:

 Even after OS reinstall → Slow

Causes:

- Old CPU
- Low RAM
- BIOS not updated

- Drivers missing
-

Solution Approach:

1. Check CPU usage
 2. Check RAM usage
 3. Update drivers
 4. Check BIOS version
-

Real IT Support Thinking

 Always think like this:

- What is the problem?
 - Hardware or Software?
 - Check basics first
 - Verify before conclusion
-

Final Understanding

 CPU = Processing

 RAM = Temporary space

 GPU = Display

 Performance depends on:

- CPU cores
 - RAM size
 - Storage speed
-

Chapter 3 (SUMMARY)

CPU (Processor – Brain of Computer)

 CPU ka kaam:

- All calculations 
- All processing 
- System control 

Core & Thread (VERY IMPORTANT)

-  Core = Physical unit
-  Thread = Virtual unit

 Concept:

- 1 Core = 1 task
- 1 Core + 2 Threads = 2 tasks

 Example:

- Core = Worker 
- Thread = Hands 

 More cores = Better multitasking 

CPU Categories

- i3 → Basic use
 - i5 → Normal use
 - i7 → Heavy work
 - i9 → Extreme performance
-

CPU Naming (Interview Important)

👉 Example: i5-8250U

- 8 → Generation
- U → Low power

Suffix Meaning:

- U → Battery saving
 - H → High performance
 - K → Overclockable
 - F → No GPU
-

RAM (Temporary Memory)

👉 RAM ka kaam:

- Running apps ka data store karna

💡 Example:

- CPU = Student 🎓
 - RAM = Study table 📚
-

RAM Types

- DDR5 → Fastest
- DDR4 → Common
- DDR3 → Old

👉 DDR = Double Data Rate

GPU (Graphics Unit)

👉 GPU ka kaam:

- Graphics process karna
- Video, gaming, rendering

💡 Example:

- CPU = Brain 🧠
- GPU = Artist 🎨

System Information Tools

msinfo32

👉 Shows:

- CPU
- RAM
- BIOS
- OS

winver

👉 Shows Windows version

Heating Concept

👉 More power = More heat

👉 Bad airflow = Overheating

❌ Avoid:

- Bed, blanket

✅ Use:

- Flat surface
-

Troubleshooting (Core Concept)

 System slow even after OS install:

- CPU old
 - RAM low
 - Drivers missing
 - BIOS outdated
-

CONCLUSION

 System performance depends on:

- CPU (Processing power) 
- RAM (Speed & multitasking) 
- GPU (Graphics performance) 

 Final line:

“Right hardware + Proper troubleshooting = Fast system” 

Q & A

◆ CPU QUESTIONS

 **Q1: What is CPU?**

 **Answer:**

CPU (Central Processing Unit) is the main component of a computer that performs all calculations and executes instructions.
It is known as the brain of the computer because it controls all operations.

? Q2: What is the difference between core and thread?

👉 Answer:

- Core = Physical processing unit
- Thread = Virtual processing path

👉 Example:

A CPU with 4 cores and 8 threads can handle 8 tasks simultaneously, improving multitasking and performance.

? Q3: Why are more cores important?

👉 Answer:

More cores allow the CPU to perform multiple tasks at the same time, improving performance in multitasking, gaming, and heavy applications.

? Q4: Explain CPU naming like i5-8250U?

👉 Answer:

- i5 → Processor category
- 8 → Generation
- U → Low power processor

👉 This helps identify performance level and power consumption.

? Q5: What are CPU suffix types?

👉 Answer:

- U → Low power
 - H → High performance
 - K → Overclockable
 - F → No integrated GPU
-

◆ RAM QUESTIONS

? Q6: What is RAM?

👉 Answer:

RAM (Random Access Memory) is temporary memory used by CPU to store data of running applications.

It is volatile, meaning data is lost when power is OFF.

? Q7: Why RAM is important?

👉 Answer:

RAM allows fast data access for running applications.

More RAM = better multitasking and smoother performance.

? Q8: What is DDR?

👉 Answer:

DDR stands for Double Data Rate.

It means data transfer speed is doubled compared to older memory types.

? Q9: Difference between DDR3, DDR4, DDR5?

👉 Answer:

- DDR3 → Slow, old
 - DDR4 → Balanced
 - DDR5 → Faster, modern
-

◆ GPU QUESTIONS

? Q10: What is GPU?

👉 Answer:

GPU (Graphics Processing Unit) is responsible for rendering images, videos, and graphics on the screen.

? Q11: Difference between CPU and GPU?

👉 Answer:

CPU 🧠	GPU 🎨
Handles logic	Handles graphics
General tasks	Parallel tasks

? Q12: When GPU is required?

👉 Answer:

- Gaming
 - Video editing
 - 3D rendering
-

◆ SYSTEM INFO QUESTIONS

? Q13: What is msinfo32?

👉 Answer:

A Windows tool used to check detailed system information like CPU, RAM, BIOS, and OS details.

? **Q14: What is winver?**

👉 **Answer:**

Command used to check Windows version.

◆ **TROUBLESHOOTING QUESTIONS (VERY IMPORTANT)**

🐢 **SYSTEM SLOW**

? **Q15: System is slow even after OS reinstall?**

👉 **Answer:**

Step-by-step:

1. Check CPU usage
 2. Check RAM usage
 3. Install drivers
 4. Update BIOS
 5. Check hardware limitations
-

? **Q16: How do you identify hardware bottleneck?**

👉 **Answer:**

- High CPU usage → CPU bottleneck
 - High RAM usage → RAM issue
 - GPU usage high → Graphics load
-

OVERHEATING

 Q17: Why system overheats?

 Answer:

- Dust in fan
 - Poor airflow
 - High CPU usage
-

 Q18: How to fix overheating?

 Answer:

- Clean fan
 - Use cooling pad
 - Ensure airflow
-

REAL JOB SCENARIOS

 Q19: Laptop heating issue?

 Answer:

- Check airflow
 - Clean fan
 - Avoid bed use
 - Replace thermal paste
-

? Q20: System hangs while multitasking?

👉 Answer:

👉 Possible causes:

- Low RAM
- Weak CPU

👉 Solution:

- Upgrade RAM
- Close background apps

? Q21: How to check system configuration?

👉 Answer:

- Use msinfo32
- Use task manager

◆ INTERVIEW IMPRESSION QUESTIONS

? Q22: What affects system performance?

👉 Answer:

- CPU cores
 - RAM size
 - Storage speed
 - GPU
-

? Q23: How do you troubleshoot system performance issue?

👉 Answer:

- Identify problem
 - Check hardware usage
 - Update drivers
 - Optimize system
-

KNM